Theoretical Quantum Optics: Problem Sheet 9
(for 14.07.2025)

26. Squeezing spectrum (2P)

In the lecture, we studied a parametrically driven cavity mode with Hamiltonian

$$H = i\frac{\hbar G}{2}\left(e^{i\theta}a_s^{\dagger 2} - e^{-i\theta}a_s^2\right) \tag{1}$$

and a decay rate γ_s.

(a) Derive the QLEs for the quadrature operators

$$X_\theta = \frac{1}{\sqrt{2}}\left(e^{i\theta}a_s^\dagger + e^{-i\theta}a_s\right), \qquad Y_\theta = \frac{i}{\sqrt{2}}\left(e^{i\theta}a_s^\dagger - e^{-i\theta}a_s\right),$$

and derive the results for $\langle \dot{X}_\theta\rangle$ and $\langle \dot{Y}_\theta\rangle$ from the lecture.

(b) Solve the QLE equations in Fourier space and derive the input-output relations

$$X_{\rm out}(\omega) = \ldots X_{\rm in}(\omega), \qquad Y_{\rm out}(\omega) = \ldots Y_{\rm in}(\omega),$$

where $X_{\rm in} = (b_{\rm in} + b_{\rm in}^\dagger)/\sqrt{2}$, etc.

(c) Evaluate the squeezing spectrum $S_{YY}(\omega)$ defined as

$$\langle Y_{\rm out}(\omega)Y_{\rm out}(\omega')\rangle = S_{YY}(\omega)\delta(\omega+\omega'). \tag{2}$$

27. Two-photon correlations for a displaced squeezed state (2P)

In the lecture, we defined a general displaced squeezed state as

$$|\alpha,\xi\rangle = D(\alpha)S(\xi)|{\rm vac}\rangle \tag{3}$$

with $D(\alpha)$ and $S(\xi)$ being the displacement operator and the squeezing operator, respectively. In this exercise, we are interested in the behavior of the second-order coherence function,

$$g^{(2)}(0) = \frac{\langle a^\dagger a^\dagger a a\rangle}{\langle a^\dagger a\rangle^2}, \tag{4}$$

for this state.

(a) Based on the definition of the squeezing operator, do you expect $g^{(2)}(0) > 1$ or $g^{(2)}(0) < 1$?

(b) Evaluate $g^{(2)}(0)$ for a squeezed vacuum state with $\xi \neq 0$ and $\alpha = 0$. What is the asymptotic behavior of $g^{(2)}(0)$ for $|\xi| \to 0$ and $|\xi| \to \infty$?

(c) Consider a weakly displaced and weakly squeezed state with $|\xi| \ll 1$ and $|\alpha| \ll 1$. Expand the squeezing and the displacement operators up to the lowest relevant orders in ξ and α and identify parameters for which $g^{(2)}(0) \approx 0$.

28. Two-mode squeezed states: stability (1P)

Consider two cavity modes that are parametrically driven and described by the master equation

$$\dot{\rho} = -i[H, \rho] + \gamma_1 \mathcal{D}[a_1]\rho \ + \gamma_2 \mathcal{D}[a_2]\rho, \tag{5}$$

where $\mathcal{D}[a]\rho \ = a\rho a^\dagger - (a^\dagger a\rho + \rho a^\dagger a)/2$ and

$$H = i\frac{\hbar G}{2}\left(a_1^\dagger a_2^\dagger - a_1 a_2\right) \tag{6}$$

is the two-mode squeezing Hamiltonian.

Derive the equations of motion for the mean-values of $a_1, a_1^\dagger, a_2, a_2^\dagger$ and determine the critical coupling strength G_c, above which the system becomes unstable. Assume $\gamma_1 \geq \gamma_2$.

29. Two-mode squeezed states: numerics (1P)

Consider the same model of as in the previous exercise with $\gamma_1 = \gamma_2 = \gamma$.

1. Write a program in Python to simulate the steady state of this system for $G < G_c$ and plot the variances of the quadratures $X_{1,2}$, $Y_{1,2}$, $X_\pm = (X_1 \pm X_2)/\sqrt{2}$ and $Y_\pm = (Y_1 \pm Y_2)/\sqrt{2}$. Verify that the steady state violates the Duan-Simon criterion, i.e., that

$$\langle X_-^2 \rangle + \langle Y_+^2 \rangle < 1. \tag{7}$$

2. The steady state of this system is expected to be close to a two-mode squeezed state of the form

$$|\Psi_{\mathrm{TMS}}\rangle \sim \sum_n \lambda^n |n\rangle_1 \otimes |n\rangle_2. \tag{8}$$

Evaluate the correlations between the number states in both modes by evaluating the correlation matrix

$$C(n,m) = \langle |n\rangle_1\langle n| \otimes |m\rangle_2\langle m| \rangle - \langle |n\rangle_1\langle n| \otimes \mathbb{1}_2\rangle\langle \mathbb{1}_1 \otimes |m\rangle_2\langle m|\rangle \tag{9}$$

for the steady state. Use a pcolor or similar plot to illustrate these correlations.

Hint: These simulations require a large Hilbertspace, especially near $G \sim G_c$. Start with small values $G \approx 0.5G_c$ and verify that the cutoff number $n_{\max}$ for each mode is sufficiently large to obtain converged numerical results. Then gradually increase the coupling, but stay below the instability.

I assume I made an error with the θ's right when deriving the QLE's but I really can't find it....

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and a decay rate γ_s.

(a) Derive the QLEs for the quadrature operators

$$X_\theta = \frac{1}{\sqrt{2}}\left(e^{i\theta}a_s^\dagger + e^{-i\theta}a_s\right), \qquad Y_\theta = \frac{i}{\sqrt{2}}\left(e^{i\theta}a_s^\dagger - e^{-i\theta}a_s\right),$$

and derive the results for $\langle \dot{X}_\theta \rangle$ and $\langle \dot{Y}_\theta \rangle$ from the lecture.

(b) Solve the QLE equations in Fourier space and derive the input-output relations

$$X_{\rm out}(\omega) = \ldots X_{\rm in}(\omega), \qquad Y_{\rm out}(\omega) = \ldots Y_{\rm in}(\omega),$$

where $X_{\rm in} = (b_{\rm in} + b_{\rm in}^\dagger)/\sqrt{2}$, etc.

(c) Evaluate the squeezing spectrum $S_{YY}(\omega)$ defined as

$$\langle Y_{\rm out}(\omega) Y_{\rm out}(\omega')\rangle = S_{YY}(\omega)\delta(\omega + \omega'). \tag{2}$$

$$\dot X_\theta = \frac{i}{\hbar}[H, X_\theta] - [X_\theta, a^\dagger]\left(\frac{\gamma_s}{2}a + \sqrt{\gamma_s}\, b_{in}(t)\right) - \left(\frac{\gamma_s}{2}a^\dagger + \sqrt{\gamma}\, b_{in}^\dagger(t)\right)[a, X_\theta]$$

$$\frac{i}{\hbar}[H, X_\theta] = -\frac{G}{2}\left[e^{i\theta}\hat a_s^{\dagger 2} - e^{-i\theta}\hat a_s^2, \frac{1}{\sqrt2}\left(e^{i\theta}a_s^\dagger + e^{-i\theta}a_s\right)\right] = -\frac{G}{2\sqrt2}\left(e^{i\theta}e^{-i\theta}[\hat a_s^{\dagger 2}, a_s] - \underbrace{e^{-i\theta}e^{i\theta}}_{=e^{i(\theta-\theta)}=1}[a_s^2, a_s^\dagger]\right)$$

$$= -\frac{G}{2\sqrt2}\left(-2\hat a^\dagger - 2a_s\right) = \frac{G}{\sqrt2}\left(a_s^\dagger + a_s\right) = G X_0$$ lecture says should be X_θ?

$$[X_0, a^\dagger] = \frac{1}{\sqrt2}e^{-i\theta}[a_s, a^\dagger] = \frac{1}{\sqrt2}e^{-i\theta}\mathbb{1}$$

$$[a, X_0] = \frac{1}{\sqrt2}e^{i\theta}\mathbb{1}$$

$\langle b_{in}(t)\rangle = 0$

$$\Rightarrow \dot X_\theta = G X_0 - \frac{1}{\sqrt2}e^{-i\theta}\left(\frac{\gamma_s}{2}a + \sqrt{\gamma_s}\, b_{in}(t)\right) - \frac{1}{\sqrt2}e^{i\theta}\left(\frac{\gamma_s}{2}\hat a^\dagger + \sqrt{\gamma_s}\, b_{in}^\dagger(t)\right)$$

$$\langle \dot X_\theta\rangle = G\langle X_0\rangle - \frac{1}{\sqrt2}\frac{\gamma_s}{2}\left(e^{-i\theta}\langle a\rangle + e^{i\theta}\langle a^\dagger\rangle\right)$$

" $G X_0 - \frac{\gamma_s}{2}X_0 - \frac{\sqrt{\gamma_s}}{\sqrt2}\left(e^{-i\theta}b_{in} + e^{i\theta}b_{in}^\dagger\right)$

Y_θ:

$$\frac{i}{\hbar}[H, Y_\theta] = -i\frac{G}{2}\left[e^{i\theta}\hat a_s^{\dagger 2} - e^{-i\theta}\hat a_s^2, \frac{1}{\sqrt2}\left(e^{i\theta}a_s^\dagger - e^{-i\theta}a_s\right)\right]$$

$$= -\frac{iG}{\sqrt2\, 2}\left(-[e^{i\theta}\hat a_s^{\dagger 2}, e^{-i\theta}\hat a_s] - [e^{-i\theta}\hat a_s^2, e^{i\theta}\hat a_s^\dagger]\right)$$

$$= -\frac{iG}{2\sqrt2}\left(2e^{i(\theta-\theta)}\hat a_s^\dagger - 2\hat a_s\right) = -\frac{iG}{\sqrt2}\left(\hat a_s^\dagger + \hat a_s\right) = -G Y_0$$ same here Y_θ

$$[Y_\theta, a^\dagger](\ldots) = -\frac{i}{\sqrt2}e^{-i\theta}\left(\frac{\gamma_s}{2}a + \sqrt{\gamma_s}\, b_{in}(t)\right)$$

$$(\ldots)[a, Y_\theta] = \frac{i}{\sqrt2}e^{i\theta}\left(\frac{\gamma_s}{2}\hat a^\dagger + \sqrt{\gamma_s}\, b_{in}^\dagger(t)\right)$$

$$\rightarrow \dot Y_\theta = -G Y_0 + \frac{i}{\sqrt2}e^{-i\theta}\left(\frac{\gamma_s}{2}a + \sqrt{\gamma_s}\, b_{in}(t)\right) - \frac{i}{\sqrt2}e^{i\theta}\left(\frac{\gamma_s}{2}\hat a^\dagger + \sqrt{\gamma_s}\, b_{in}^\dagger(t)\right)$$

$$\rightarrow \langle \dot Y_\theta\rangle = -G\langle Y_0\rangle + \frac{i}{\sqrt2}\frac{\gamma_s}{2}\left(e^{-i\theta}\langle a\rangle - e^{i\theta}\langle a^\dagger\rangle\right)$$

b)

Fourier

$$\dot{X}_\theta = G X_\theta - \frac{1}{\sqrt{2}} e^{-i\theta}\left(\frac{\gamma_S}{2} a + \sqrt{\gamma_S}\, b_{in}(t)\right) - \frac{1}{\sqrt{2}} e^{i\theta}\left(\frac{\gamma_S}{2} \hat{a}^+ + \sqrt{\gamma_S}\, b_{in}^+(t)\right)$$

$\downarrow$

$$-i\omega X_\theta = G X_\theta(\omega) - \frac{\gamma_S}{2} X_\theta(\omega) - \sqrt{\frac{\gamma_S}{2}}\left(e^{-i\theta} b_{in}(\omega) + e^{i\theta} b_{in}^+(\omega)\right)$$

$$-i\omega Y_\theta = -G Y_\theta(\omega) - \frac{\gamma_S}{2} Y_\theta(\omega) - i\sqrt{\frac{\gamma_S}{2}}\left(e^{i\theta} b_{in}^+(\omega) - e^{-i\theta} b_{in}(\omega)\right)$$

$$X_{out} \overset{?}{=} \frac{b_{out} + b_{out}^+}{\sqrt{2}} = \frac{b_{in} + b_{in}^+}{\sqrt{2}} + \frac{\sqrt{\gamma}(a + a^+)}{\sqrt{2}} = X_{in} + \sqrt{\gamma} X_0$$

$$\to \quad X_\theta = \left(\frac{iG}{\omega} - \frac{i\gamma_S}{2\omega} X_\theta(\omega)\right) - \frac{i}{\omega}\sqrt{\frac{\gamma_S}{2}}\left(e^{-i\theta} b_{in}(\omega) + e^{i\theta} b_{in}^+(\omega)\right)$$

$$X_\theta(\omega)\underbrace{\left(1 - \frac{iG}{\omega} + \frac{i\gamma_S}{2\omega}\right)}_{K} = \frac{-i}{\omega}\sqrt{\frac{\gamma_S}{2}}\left(e^{-i\theta} b_{in}(\omega) + e^{i\theta} b_{in}^+(\omega)\right)$$

$$X_\theta(\omega) = \frac{-i}{K\omega}\sqrt{\frac{\gamma_S}{2}}\left(e^{-i\theta} b_{in}(\omega) + e^{i\theta} b_{in}^+(\omega)\right)$$

$$\to X_{out}(\omega) = X_{in}(\omega) - \frac{i}{K\omega}\frac{\gamma_S}{\sqrt{2}}\left(e^{-i\theta} b_{in}(\omega) + e^{i\theta} b_{in}^+(\omega)\right)$$

$$\underset{\theta=0}{=} X_{in}(\omega) - \frac{i}{K\omega}\frac{\gamma_S}{\sqrt{2}} X_{in} = \left(1 - \frac{i}{K\omega}\frac{\gamma_S}{\sqrt{2}}\right) X_{in}$$

analogously for $Y_{out}(\omega) = Y_{in} + \sqrt{\gamma}\, Y_\theta$

$$-i\omega Y_\theta = -G Y_\theta(\omega) - \frac{\gamma_S}{2} Y_\theta(\omega) - i\sqrt{\frac{\gamma_S}{2}}\left(e^{i\theta} b_{in}^+(\omega) - e^{-i\theta} b_{in}(\omega)\right)$$

$$i\omega Y_\theta(\omega) - G Y_\theta(\omega) - \frac{\gamma_S}{2} Y_\theta(\omega) = -i\sqrt{\frac{\gamma_S}{2}}\left(e^{i\theta} b_{in}^+(\omega) - e^{-i\theta} b_{in}(\omega)\right)$$

$$Y_\theta(\omega)\underbrace{\left(i\omega - G - \frac{\gamma_S}{2}\right)}_{K_Y} = -i\sqrt{\frac{\gamma_S}{2}}\left(e^{i\theta} b_{in}^+(\omega) - e^{-i\theta} b_{in}(\omega)\right)$$

$$Y_\theta(\omega) = -\frac{i}{K_Y}\frac{\sqrt{\gamma_S}}{\sqrt{2}}\left(e^{i\theta} b_{in}^+(\omega) - e^{-i\theta} b_{in}(\omega)\right)$$

$$Y_{out}(\omega) = Y_{in}(\omega) - \frac{i}{K_Y}\frac{\gamma_S}{\sqrt{2}}\left(e^{i\theta} b_{in}^+(\omega) - e^{-i\theta} b_{in}(\omega)\right)$$

$$\| \; \theta = 0$$

$$Y_{in}(\omega) - \frac{\gamma_S}{K_y} Y_{in}(\omega) = \left(1 - \frac{\gamma_S}{K_y}\right) Y_{in}(\omega)$$

c) For coherent states $\langle a^+a^+aa\rangle = |\alpha|^4 \to g^{(2)}(0) = 1$

here $\langle a^+a^+aa\rangle = \langle\alpha|\, \xi^+ a^+a^+aa\, \xi\, |\alpha\rangle$

~~the squeezing amplifies vacuum fluctuations (a qm effect), so if the squeezing parameter is large enough, I would expect~~

the state is a displaced and squeezed vaccuum. This changes $\langle X^2\rangle$, but $\langle a^+a\rangle$ and $\langle a^+a^+aa\rangle$ should not change, therefore \

would expect $g^{(2)}(0)=1$ as before the squeezing.

b)

$$S^\dagger(\xi)\, a\, S(\xi) = a\cosh(r) - a^\dagger e^{2i\phi}\sinh(r)$$
$$S^\dagger(\xi)\, a^\dagger S(\xi) = a^\dagger\cosh(r) - a\, e^{-2i\phi}\sinh(r)$$

$$g^{(2)}(0) = \frac{\langle a^\dagger a^\dagger a a\rangle}{\langle a^\dagger a\rangle} \qquad [a,a^\dagger] = \mathbb{1} \Rightarrow a a^\dagger = \mathbb{1} + a^\dagger a$$

$$\langle a^\dagger a\rangle = \langle \xi | a^\dagger a|\xi\rangle = \langle\alpha|S^\dagger a^\dagger S S^\dagger a S|\alpha\rangle = \langle\alpha|(a^\dagger\cosh(r) - a e^{-2i\phi}\sinh(r))(a\cosh(r) - a^\dagger e^{2i\phi}\sinh(r))|\alpha\rangle$$

$$= \langle\alpha|\, a^\dagger a\cosh^2(r) + a a^\dagger \sinh^2(r)|\alpha\rangle = \langle\alpha|a^\dagger a|\alpha\rangle\cosh^2(r) + \langle\alpha|\alpha\rangle\sinh^2(r) + \langle\alpha|a^\dagger a|\alpha\rangle\sinh^2(r)$$

$$= |\alpha|^2\cosh^2(r) + \sinh^2(r) + |\alpha|^2\underbrace{\sinh^2(r)}_{=2\cosh^2(r)-1} = |\alpha|^2(3\cosh^3(r) - 1) + \sinh^2(r)$$

$$\Rightarrow \langle a^\dagger a\rangle^2 = \left(|\alpha|^2\cosh^2(r) + \sinh^2(r) + |\alpha|^2\sinh^2(r)\right)^2 = \left(|\alpha|^2(\cosh^2(r)+\sinh^2(r)) + \sinh^2(r)\right)^2 = |\alpha|^4(\cosh^2(r)+\sinh^2(r))^2 + \sinh^4(r) + 2|\alpha|^2\sinh^2(r)(\cosh^2(r)+\sinh^2(r))$$

$$[aa, a^\dagger] = a[a,a^\dagger] = -a$$
$$[aa, a^\dagger] = -a$$
$$aaa^\dagger = a^\dagger aa - a$$

$$\langle a^\dagger a^\dagger a a\rangle^2 = \langle\alpha|S^\dagger a^\dagger S S^\dagger a^\dagger S S^\dagger a S S^\dagger a S|\alpha\rangle$$

$$= \langle\alpha|(a^\dagger\cosh(r) - a e^{-2i\phi}\sinh(r))^2 (a\cosh(r) - a^\dagger e^{2i\phi}\sinh(r))^2|\alpha\rangle$$

$$= \langle\alpha|(a^{\dagger 2}\cosh^2(r) + a^2 e^{-4i\phi}\sinh^2(r) - a^\dagger a\, e^{-2i\phi}\cosh(r)\sinh(r) - a a^\dagger e^{-2i\phi}\cosh(r)\sinh(r))(a^2\cosh^2(r) + a^{\dagger 2} e^{4i\phi}\sinh^2(r)$$
$$- a a^\dagger\cosh(r)\sinh(r) e^{2i\phi} - a^\dagger a\, e^{2i\phi}\cosh(r)\sinh(r)|\alpha\rangle$$

$$= \cosh^4(r)\langle a^{\dagger 2} a^2\rangle + \sinh^4(r)\langle a^2 a^{\dagger 2}\rangle + \cosh^2(r)\sinh^2(r)\left(\langle a^\dagger a a a^\dagger\rangle + \langle a^\dagger a a^\dagger a\rangle + \langle a a^\dagger a a^\dagger\rangle + \langle a a^\dagger a^\dagger a\rangle\right)$$

$$= |\alpha|^4(\cosh^4(r) + \sinh^4(r)) + \cosh^2(r)\sinh^2(r)\big(\langle a^\dagger(a^\dagger a a - a)\rangle + \langle a^\dagger(a^\dagger a + \mathbb{1})a\rangle + \langle(a^\dagger a + \mathbb{1})(a^\dagger a + \mathbb{1})\rangle$$
$$+ \langle(a^\dagger a + \mathbb{1})a^\dagger a\rangle\big)$$

$$= |\alpha|^4(\cosh^4(r) + \sinh^4(r)) + \cosh^2(r)\sinh^2(r)\left(\langle a^\dagger a^\dagger a a\rangle - \langle a^\dagger a\rangle + \langle a^\dagger a^\dagger a a\rangle + \langle a^\dagger a\rangle + \langle a^\dagger a a^\dagger a\rangle + \langle\mathbb{1}\rangle + 2\langle a^\dagger a\rangle + \langle a^\dagger a a^\dagger a\rangle + \langle a^\dagger a\rangle\right)$$

$$= |\alpha|^4(\cosh^4(r) + \sinh^4(r)) + \cosh^2(r)\sinh^2(r)\big(|\alpha|^4 - |\alpha|^2 + |\alpha|^4 + |\alpha|^2 + |\alpha|^4 + 1 + 2|\alpha|^2 + \overbrace{\langle a^\dagger a^\dagger a a\rangle}^{=|\alpha|^4} + \overbrace{\langle a^\dagger a\rangle + \langle a^\dagger a\rangle}^{=2|\alpha|^2}\big)$$

$$= |\alpha|^4(\cosh^4(r) + \sinh^4(r)) + \cosh^2(r)\sinh^2(r)(4|\alpha|^4 + 4|\alpha|^2 + 1) = |\alpha|^4(\cosh^4(r) + \sinh^4(r)) + \cosh^2(r)\sinh^2(r)(2|\alpha|^2 + 1)^2$$

$$\Rightarrow g^{(2)}(0) = \frac{|\alpha|^4(\cosh^4(r) + \sinh^4(r)) + \cosh^2(r)\sinh^2(r)(2|\alpha|^2+1)^2}{\left(|\alpha|^2\cosh^2(r) + \sinh^2(r) + |\alpha|^2\sinh^2(r)\right)^2}$$

<u>if $\alpha \neq 0$</u>

for $|\xi| \to 0$: $r \to 0 \Rightarrow \cosh(r) \to 1$, $\sinh(r) \to 0$

$$\rightarrow \frac{|\alpha|^4(\cosh^4(r) + \sinh^4(r)) + \cosh^2(r)\sinh^2(r)(2|\alpha|^2+1)^2}{\left(|\alpha|^2\cosh^2(r) + \sinh^2(r) + |\alpha|^2\sinh^2(r)\right)^2} \rightarrow \frac{|\alpha|^4(1+0) + 1\cdot 0}{(|\alpha|^2\cdot 1 + 0)^2} = \frac{|\alpha|^4}{|\alpha|^4} = 1$$

for $|\xi| \to \infty$: $r \to \infty \Rightarrow \cosh(r), \sinh(r) \to \infty$, $g^{2}(0) \to \infty$ (I forgot that we had to take $\alpha = 0$ and now that I've already calculated it it's nice to see how it behaves ☺)

now for $|\alpha|=0$:

$$\frac{|\alpha|^4\left(\cosh^4(r)+\sinh^4(r)\right)+\cosh^2(r)\sinh^2(r)\left(2|\alpha|^2+1\right)^2}{\left(|\alpha|^2\cosh^2(r)+\sinh^2(r)+|\alpha|^2\sinh^2(r)\right)^2} = \frac{\cosh^2(r)\sinh^2(r)}{\sinh^4(r)} = \frac{\cosh^2(r)}{\sinh^2(r)} = (*)$$

for $r\to 0$: $(*) \to$ „$\frac{1}{0}$" $\to \infty$

$r\to\infty$: $\sinh(x)\to e^x,\ \cosh(x)\to e^x \Rightarrow (*)\to 1$ //

c)

$$\frac{|\alpha|^4\left(\cosh^4(r)+\sinh^4(r)\right)+\cosh^2(r)\sinh^2(r)\left(2|\alpha|^2+1\right)^2}{\left(|\alpha|^2\cosh^2(r)+\sinh^2(r)+|\alpha|^2\sinh^2(r)\right)^2}$$

lowest relevant orders: $|\alpha|^2=|\alpha|^2$
$|\alpha|^4 \approx 0$

$$\cosh(r) = 1+\frac{r^2}{2}+\frac{r^4}{4!}+\dots \approx 1+\frac{r^2}{2}$$
$$\sinh(r) = r+\frac{r^3}{3!}+\frac{r^5}{5!}+\dots \approx r$$

$$= \frac{\left(1+\frac{r^2}{2}\right)^2\cdot r^2\left(2|\alpha|^2+1\right)^2}{\left(|\alpha|^2(1+r^2)+r^2+|\alpha|^2r^2\right)^2} + \mathcal{O}(r^3,\alpha^3)$$

$$= \frac{(1+r^2)\cdot r^2\left(4|\alpha|^2+1\right)}{\left(|\alpha|^2(1+r^2)+r^2(1+|\alpha|^2)\right)^2} + \mathcal{O}(r^3,\alpha^3) = \frac{r^2\left(4|\alpha|^2+1\right)}{2|\alpha|^2(1+r^2)\,r^2(1+|\alpha|^2)} + \mathcal{O}(r^3,\alpha^3)$$

$$\approx \frac{4|\alpha|^2+1}{2|\alpha|^2(1+r^2)(1+|\alpha|^2)} = \frac{2}{(1+r^2)(1+|\alpha|^2)} + \frac{1}{2|\alpha|^2} \to \infty \Rightarrow g^{(2)}(0)\approx 0 \text{ never works (?)}$$

28.

$$\dot{\rho} = -\frac{i}{\hbar}[H,\rho] + \sum_k \frac{\Gamma_k}{2}(2c_k\rho c_k^\dagger - c_k^\dagger c_k\rho - \rho c_k^\dagger c_k)$$

$$\langle O\rangle(t) = \mathrm{Tr}(O\rho(t)) \rightarrow \langle\dot{O}\rangle(t) = \mathrm{Tr}(O\dot{\rho}(t)) = \frac{i}{\hbar}\langle[H,O]\rangle + \sum_k \frac{\Gamma_k}{2}\langle[c_k^\dagger,O]c_k + c_k^\dagger[O,c_k]\rangle$$

with $c_1 = a_1$, $\Gamma_1 = \gamma_1$
$c_2 = a_2$, $\Gamma_2 = \gamma_2$

$\langle a_1\rangle$: $\langle\dot{a}_1\rangle(t) = \frac{i}{\hbar}\langle[H,a_1]\rangle + \frac{\gamma_1}{2}\langle[a_1^\dagger,a_1]a_1 + a_1^\dagger[a_1,a_1]\rangle + \frac{\gamma_2}{2}\cdot 0$

$$[H,a_1] = \frac{i\hbar G}{2}[a_1^\dagger,a_1]a_2^\dagger = -\frac{G}{2}\cdot(-1)\hat{a}_2^\dagger = \frac{G}{2}\hat{a}_2^\dagger$$

$$\rightarrow \langle\dot{a}_1\rangle(t) = \frac{G}{2}\langle\hat{a}_2^\dagger\rangle - \frac{\gamma_1}{2}\langle\hat{a}_1\rangle$$

$$\langle\dot{a}_2\rangle(t) = \frac{i}{\hbar}\langle[H,a_2]\rangle + \frac{\gamma_2}{2}\langle[a_2^\dagger,a_2]a_2 + a_2^\dagger[a_2,a_2]\rangle = \frac{G}{2}\langle\hat{a}_1^\dagger\rangle - \frac{\gamma_2}{2}\langle\hat{a}_2\rangle$$

$$\langle\dot{a}_1^\dagger\rangle(t) = \frac{i}{\hbar}\langle[H,a_1^\dagger]\rangle + \frac{\gamma_1}{2}a_1^\dagger\langle[a_1^\dagger,a_1]\rangle = \frac{G}{2}\langle\hat{a}_2\rangle - \frac{\gamma_1}{2}\langle a_1^\dagger\rangle$$

$$\frac{i}{\hbar}\langle[H,a_1^\dagger]\rangle = +\frac{G}{2}[a_1,a_1^\dagger]a_2 = \frac{G}{2}\hat{a}_2$$

$$\langle\dot{a}_2^\dagger\rangle(t) = \frac{G}{2}\langle\hat{a}_1\rangle - \frac{\gamma_2}{2}\langle\hat{a}_2^\dagger\rangle$$

$$\Rightarrow \quad \langle\dot{a}_1\rangle(t) = \frac{G}{2}\langle\hat{a}_2^\dagger\rangle - \frac{\gamma_1}{2}\langle\hat{a}_1\rangle$$
$$\langle\dot{a}_2\rangle(t) = \frac{G}{2}\langle\hat{a}_1^\dagger\rangle - \frac{\gamma_2}{2}\langle\hat{a}_2\rangle$$
$$\langle\dot{a}_1^\dagger\rangle(t) = \frac{G}{2}\langle\hat{a}_2\rangle - \frac{\gamma_1}{2}\langle a_1^\dagger\rangle$$
$$\langle\dot{a}_2^\dagger\rangle(t) = \frac{G}{2}\langle\hat{a}_1\rangle - \frac{\gamma_2}{2}\langle\hat{a}_2^\dagger\rangle$$

$$\rightarrow \begin{pmatrix}\langle a_1\rangle\\ \langle a_2\rangle\\ \langle a_1^\dagger\rangle\\ \langle a_2^\dagger\rangle\end{pmatrix} = \begin{pmatrix}-\gamma_1 & 0 & 0 & G\\ 0 & -\gamma_2 & G & 0\\ 0 & G & -\gamma_1 & 0\\ G & 0 & 0 & -\gamma_2\end{pmatrix}\begin{pmatrix}\langle a_1\rangle\\ \langle a_2\rangle\\ \langle a_1^\dagger\rangle\\ \langle a_2^\dagger\rangle\end{pmatrix}$$

$$\lambda_{1,2} = -\frac{\gamma_1+\gamma_2}{4} \pm \frac{1}{4}\underbrace{\sqrt{(\gamma_1-\gamma_2)^2+4G^2}}_{=K}$$

$$\lambda_{3,4} = \lambda_{1,2}$$

$$\vec{v}_1 = \begin{pmatrix}1\\0\\0\\ \frac{\gamma_1+2\lambda_1}{G}\end{pmatrix} \quad \vec{v}_2 = \begin{pmatrix}1\\0\\0\\ \frac{\gamma_1+2\lambda_2}{G}\end{pmatrix}$$

$$\vec{v}_3 = \begin{pmatrix}0\\1\\ \frac{\gamma_2+2\lambda_1}{G}\\0\end{pmatrix} \quad \vec{v}_3 = \begin{pmatrix}0\\1\\ \frac{\gamma_2+2\lambda_2}{G}\\0\end{pmatrix}$$

$$\Rightarrow \begin{pmatrix}\langle a_i\rangle(t)\\ \langle a_j^\dagger\rangle(t)\end{pmatrix} = C_1 e^{\lambda_1 t}\begin{pmatrix}1\\ (\gamma_i + \frac{2\lambda_1}{G})\end{pmatrix} + C_2 e^{\lambda_2 t}\begin{pmatrix}1\\ +(\gamma_i + \frac{2\lambda_2}{G})\end{pmatrix}, \quad i \neq j$$

$$\Rightarrow \langle a_i\rangle(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} \qquad i\neq j$$
$$\langle a_j^\dagger\rangle(t) = C_1 e^{\lambda_1 t}(\gamma_i + \frac{2\lambda_1}{G}) + C_2 e^{\lambda_2 t}(\gamma_i + \frac{2\lambda_2}{G})$$

instability, when λ_i has a real part >0

$$\lambda_\pm = -\frac{\gamma_1+\gamma_2}{4} \pm \frac{1}{4}\underbrace{\sqrt{(\gamma_1-\gamma_2)^2+4G^2}}$$

λ_-: for any G it remains stable

λ_+: $-\frac{\gamma_2+\gamma_1}{4} + \frac{1}{4}\sqrt{(\gamma_1-\gamma_2)^2+4G^2} = 0$

$\gamma_2+\gamma_1 = \sqrt{(\gamma_1-\gamma_2)^2+4G^2}$

$(\gamma_2+\gamma_1)^2 = (\gamma_1-\gamma_2)^2+4G^2$

$(\gamma_2+\gamma_1)^2 - (\gamma_1-\gamma_2)^2 = 4G^2$

$2\gamma_1\gamma_2 + 2\gamma_1\gamma_2 = 4G^2$

$4\gamma_1\gamma_2 = 4G^2$

$\rightarrow$ for $G > \sqrt{\gamma_1\gamma_2}$, the system becomes unstable